NagaForge Demo Workspace

Structural Calculation Sheet

06 Jul 2026 07:33

APPROVED

Beam B12 - Level 3

Project & Basis

Calculator	**RCC Beam - IS 456:2000**
Project	**Riverside Tower - Demo**
Country / Code	**India | IS 456:2000**
Result	**-**

Design Inputs

span m	**6**
Mu kNm	**150**
Vu kN	**120**
b mm	**300**
fck	**25**
fy	**415**

Section

b mm	**300**
D mm	**350**
d mm	**309**
cover mm	**25**

Moment Capacity

Mu lim kNm	**98.64**
applied Mu kNm	**150**
is singly	**No**

Tension Steel

Ast required mm2	**1651.3**
Ast min mm2	**189.9**
bars	**9 - #16**
Ast provided mm2	**1809.9**
pt percent	**1.952**

Compression Steel

required	**Yes**
Asc mm2	**580.1**
bars	**3 - #16**

Shear Design

tau v N mm2	**1.294**
tau c N mm2	**0.811**
tau c max N mm2	**3.1**
mode	**Stirrups required**
stirrups	**2L - #8 @ 250 c/c**

stirrup spacing mm	**250**
warning	

Deflection

basic span d ratio	**20**
d provided mm	**309**
d required mm	**300**
ok	**Yes**

Development Length

Ld mm	**645**
bar dia mm	**16**

Summary

design type	**Doubly Reinforced**
beam type	**simply_supported**
code	**IS 456:2000**

Proof & Official Sources

Code Basis	**IS 456:2000**
Maturity	**Preliminary design calculator**

Clauses / tables: IS 456 flexure; IS 456 shear; IS 456 deflection; IS 456 detailing

Bureau of Indian Standards (BIS) - Government of India / National Standards Body

https://www.bis.gov.in/

Assumptions: Preliminary RCC beam design. Verify ductile detailing, development length, torsion, serviceability, exposure, and final bar arrangement.

Formula trace and official source link included; requires project review by a licensed engineer before construction.

Building law is adopted by the project jurisdiction. Verify local amendments, national annexes, authority requirements, and the enforced edition.

Certification & Sign-off

Prepared	Demo User	06 Jul 2026 02:03
Checked	R. Krishnan (Checker)	06 Jul 2026 02:03
Approved	S. Mehta, P.E. (Approver)	06 Jul 2026 02:03

This is a preliminary engineering calculation. Figures must be independently verified and the sheet sealed by the responsible licensed engineer under the governing jurisdiction before use in construction.